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COMPLIANCE MATRIX

Thesis Title : DESIGN AND IMPLEMENTATION OF A REAL-TIME LAN-BASED QUEUEING MANAGEMENT AND STATISTICAL REPORTING SYSTEM FOR THE BUSINESS PERMITS, LICENSING - LOCAL INVESTMENT AND ENTERPRISE OFFICE (BPL-LIEO) OF LGU ROSARIO, LA UNION

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Type of Examination : ☒ Proposal ☐ Final Date of Examination: **10/9/2026**

No.	Recommendations	Location	Old Page	New Page	By	Action Taken	Remarks
1	Web-Based Queuing Management System with Predictive Waiting Mechanism and Adaptive Counter Allocation	Title	i, ii, x, 3-4, 9, 19, 30, 34, 36, 38, 40, 42	i, ii, x, 3-4, 10, 28, 42, 46, 48, 50, 52, 54	ALL	*Revised the project title across the Title Page, Approval Sheet, Abstract, Chapter 1 (Situation Analysis & General Objective), Chapter 2 (Research Design & Data Analysis), Appendix C questionnaire instructions, and the researchers' Curriculum Vitae to:* ***"Web-Based Queuing Management System with Predictive Waiting Mechanism and Adaptive Counter Allocation for the BPL-LIEO of LGU Rosario, La Union"*** *to align with the panel's recommendation highlighting the predictive waiting model and adaptive counter allocation algorithm. (Signed Appendix A and Appendix D were retained as originally signed).*	Complied
2	1. To design and develop … 2. To test and improve the systems, performance of real – time cloud data synchronization and over all functionality 3. To assess the level of quality of the developed web-based system using ISO / IEC 25010 standards	SOO	4-5	4-5	ALL	*Restructured the Statement of Objectives to strictly follow the panel's mandated three-tier framework: (1) Updated General Objective and Objective 1 to encompass the design and development of the system and its core modules incorporating the Predictive Waiting Mechanism (Phantom Simulation) and Adaptive Counter Allocation Engine; (2) Refined Objective 2 to focus on testing and improving the system's performance, real-time cloud data synchronization, and overall functionality through alpha and beta testing; and (3) Realigned Objective 3 to evaluate the software quality across the ISO/IEC 25010 characteristics.*	Complied
3	Specify what particular feature you will be testing	Chapter 2	16-17	20-22	APB, CAV	*Expanded Chapter 2 under "System Testing Procedures" by replacing generic examples with an exhaustive enumeration of specific testable features mapped to Specific Objective 2, namely: (1) Ticket generation sequence integrity and QR encoding; (2) Predictive wait-time calculation via Phantom Queue Simulation evaluated through Mean Absolute Error (MAE); (3) Dynamic Smart Score ranking and Adaptive Counter Allocation mode switching (Peace vs. Crunch modes); (4) Real-time WebSocket latency (<500ms) and local-to-cloud (SQLite-to-Firestore) synchronization with offline*	Complied

						failover; (5) Automated SMS and Web Push notification delivery triggers; and (6) Staff dashboard controls and public TV display audio-visual synchronization.	
4	Include the inclusion and exclusion criteria for your participants of the study. State the reason why it is included and excluded	Chapter 2	18-19	24-28	APB	*Incorporated a comprehensive "Inclusion and Exclusion Criteria" subsection under Chapter 2 (Population and Sampling) preceding Table 1. Formulated distinct inclusion and exclusion rules along with methodological justifications for all three participant groups: (1) BPL-LIEO frontline staff (including only personnel actively operating permit counters; excluding staff from other municipal offices); (2) Transacting citizens (including legal-age clients across Regular, PWD, Senior Citizen, and Returning categories who received a ticket and observed displays/alerts; excluding non-transacting companions, minors, and clients of other departments); and (3) IT Experts (requiring formal computing credentials and software industry background to evaluate ISO/IEC 25010 technical characteristics; excluding non-technical individuals and project proponents).*	Complied
5	Maximum of 150 words. Focus on Chapter 1 and 2 only	Abstract	x	x	APB	*Condensed the abstract from 225 words to exactly 137 words, strictly conforming to the 150-word maximum requirement. The content was refocused exclusively on the problem context, proposed system, and specific objectives (Chapter 1), followed by the Agile SDLC framework, alpha/beta testing procedures, and ISO/IEC 25010 evaluation methodology with defined respondent groups (Chapter 2), deliberately omitting any preliminary results or conclusions.*	Complied
6	Include Ethical Considerations Explain your data management plan, storage of data, retention of data, and disposal of the data	Ethical Consideration	N/A	30-33	APB	*Incorporated a dedicated "Ethical Considerations" section at the conclusion of Chapter 2 adhering to Republic Act No. 10173 (Data Privacy Act of 2012) and university research ethics. Established a formal Data Management Plan detailing: (1) Voluntary informed consent and anonymity for survey respondents; (2) Data collection boundaries (restricting inputs to queue tickets and optional phone numbers, excluding financial data); (3) Secure on-premise and encrypted cloud storage utilizing bcrypt password hashing and JWT access controls; (4) A defined retention policy where citizen contact numbers for SMS alerts are purged/masked daily upon transaction completion while de-identified transaction logs are retained for 1 fiscal year for administrative reporting; and (5) Secure digital truncation and physical document shredding for data disposal.*	Complied
7	Align based on SOO No.3	Chapter 2	18-20	22-24, 28-29		*Harmonized Chapter 2 with Specific Objective No. 3 by: (1) Correcting the narrative count from seven (7) to all eight (8) ISO/IEC 25010 quality characteristics, formally incorporating Compatibility (co-existence and interoperability) alongside*	Complied

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						Functional Suitability and Performance Efficiency; (2) Structuring respondent evaluation scopes so that frontline staff and citizens assess user-facing dimensions (Functional Suitability, Usability, Reliability, Performance Efficiency) while IT experts evaluate both user-facing and technical dimensions (Security, Maintainability, Compatibility, Portability); and (3) Aligning the Data Analysis section to compute individual weighted means per ISO characteristic and an overall composite mean to directly satisfy Objective 3.	
8	Check your format, include the time of downloading	References	22–25	34–37	APB	*Reviewed and standardized all bibliographic entries under Literature Cited in accordance with institutional guidelines. Updated all online web sources, institutional portals, and agency news articles to include full retrieval metadata specifying both the access date and the time of downloading (e.g., "Retrieved October 4, 2026, 14:15, from [URL]"). Cleaned typographical symbol artifacts and normalized standard HTTPS DOI links for academic journal articles.*	Complied

Expected System Features/ Output	
1	Complied. Fully formulated and documented the three mathematical models in Chapter 2 (Pages 11–14): (1) The Dynamic Smart Score priority ranking algorithm $S(t)$; (2) The Phantom Queue Simulation wait-time prediction model $E[W]$with dynamic abandonment discounting D; and (3) The Adaptive Counter Allocation engine with proportional load balancing between Peace and Crunch modes, starvation floor enforcement, and hysteresis deadband.
2	Complied. Documented the automated multi-channel notification module in Chapter 1 (SOO 1.5) and Chapter 2 (Materials & Testing Procedures), detailing the programmatic dispatch of SMS alerts via SMS Gateway API across three automated triggers (Near-Turn Alert when $R_{true} \leq 2$, Initial Call Alert with Window number, and Recall Alert).
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This is to certify that the aforementioned recommendations were made and agreed upon during the defense by the researchers and the Thesis Committee composed of:

CHARLIE S. MARZAN
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